

Erdős Problem #963

EP-963

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $f(n)$ be the maximal k such that in any set $A \subset \mathbb{R}$ of size n there is a subset $B \subseteq A$ of size $|B| \geq k$ which is dissociated that is, the sums $\sum_{b \in S} b$ are distinct for all $S \subseteq B$. Estimate $f(n)$ - in particular, is it true that $f(n) \geq \lfloor \log_2 n \rfloor$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.
Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>